The direct method.

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We will describe how we can determine minimal periods of symmetric shift registers by a direct method for all types of start strings. The direct method is based on the definition of the given symmetric shift register.

The direct method works well if the periods are less than about 10 000 000. But if the periods are larger, then this approach is time consuming. If the period is very large, the direct approach is impossible. Then the indirect method in menu option 3 can be used. The results and methods are based on [1] "Representation of symmetric shift registers" by Jan Søreng is published and available at arXiv [2505.23974].

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1. Notation.

If $j \geq 1$, then 0_j denotes j consecutive 0's, and 1_j denotes j consecutive 1's. Moreover, if $j = 0$, then 0_j and 1_j denote the empty string.

We call $a_1 \cdots a_n$ a string if $a_i \in \{0, 1\}$ for $1 \leq i \leq n$. In the same way we also call $a_1 a_2 \cdots$ a string if $a_i \in \{0, 1\}$ for $i \geq 1$.

We suppose $A = a_1 \cdots a_n$ is a string. The weight of A is

$$w(A) = a_1 + \cdots + a_n = \text{the number of bits equal to 1 in } A.$$

2. Symmetric shift registers.

Let k, p and n be integers such that $0 \leq k \leq k + p < n$. The symmetric shift register $\theta : \{0, 1\}^n \to \{0, 1\}^n$ with respect to k, p and n is defined by $\theta(a_1 \cdots a_n) = a_2 \cdots a_{n+1}$ where

$$a_{n+1} = a_1' \text{ if } k \leq a_2 + \cdots + a_n \leq k + p, \text{ and } a_{n+1} = a_1 \text{ otherwise,}$$

where $a_1' = 0$ if $a_1 = 1$, and $a_1' = 1$ if $a_1 = 0$.

We let $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ be the string generated from A by the symmetric shift register with parameters k, p and n. For $i \geq 0$, then

$a_{n+i+1} = a_{i+1}'$ if $k \leq a_{i+2} + \cdots + a_{i+n} \leq k + p$. Otherwise, $a_{n+i+1} = a_{i+1}$.

The minimal period of A^∞ is called the minimal period of A with respect to the symmetric shift register with parameters k, p and n.

3. Vector representations.

Suppose $Q = (q_1, \cdots, q_s)$ where s is even, $q_1 \geq 0$, $q_i > 0$ for $1 \leq i < s$, and $q_s \geq 0$. Then Q is the even vector representation of the string

$$A = 1_{q_1} 0_{q_2} 1_{q_3} 0_{q_4} \cdots 1_{q_{s-1}} 0_{q_s}. \tag{3.1}$$

Example 3.1. If $A = 110011100$, then we decompose $A = 1_2 0_2 1_3 0_2$ and get that $Q = (2, 2, 3, 2)$ is the even vector representation of A.

Example 3.2. If $A = 111100111$, then we decompose $A = 1_4 0_2 1_3 0_0$ and get that $Q = (4, 2, 3, 0)$ is the even vector representation of A.

Example 3.3. If $Q = (3, 4, 2, 0)$, then Q represents

$$A = 1_3 0_4 1_2 0_0 = 111000011.$$

Example 3.4. If $Q = (2, 3, 1, 3)$, then Q represents

$$A = 1_2 0_3 1_1 0_3 = 110001000.$$

Example 3.5. If $Q = (0, 3, 1, 0)$, then Q represents $A = 1_0 0_3 1_1 0_0 = 0001$.

Example 3.6. If $Q = (0, 3, 1, 5)$, then Q represents

$$A = 1_0 0_3 1_1 0_5 = 000100000.$$

Example 3.7. If $Q = (5, 3, 1, 5, 1, 2)$, then A represents

$$A = 1_5 0_3 1_1 0_5 1_1 0_2 = 11111000100000100.$$

4. Start conditions.

We suppose the vector $Q = (q_1, \cdots, q_s)$ satisfies

$$s \text{ is even, } q_1 \geq 0, \; q_i > 0 \text{ for } 2 \leq i < s, \text{ and } q_s \geq 0.$$

Let $n = q_1 + q_2 + \cdots + q_s$ and $w = q_1 + q_3 + \cdots + q_{s-1}$.

Suppose p satisfies $0 \leq p < n$. Finally, we suppose k satisfies

$$k_{min} \leq k \leq k_{max} \tag{4.1}$$

where k_{min} and k_{max} are given by

$$k_{min} = max\{0, w - (p+1\},$$
$$k_{max} = min\{w, n - (p+1)\}.$$

Let $A = a_1 \cdots a_n$ be the string corresponding to $Q = (q_1, \cdots, q_s)$. Then

$$A = 1_{q_1} 0_{q_2} 1_{q_3} 0_{q_4} \cdots 1_{q_{s-1}} 0_{q_s}.$$

The condition (4.1) ensures that

$$0 \leq k \leq w(A) \leq k + p + 1 \leq n \text{ where } w(A) \text{ is the weight of } A.$$

Suppose $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ is generated from $A = a_1 \cdots a_n$ by the symmetric shift register with parameters k, p and n.

The goal is to determine the minimal period of A with respect to the symmetric shift register with parameters k, p and n. That means, to determine the minimal period of $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$.

5. Additional notation.

Let

$$w_i = a_{i+1} + \cdots + a_{i+n} - k \text{ for } i \geq 0. \tag{5.1}$$

Then $a_{i+1} + \cdots + a_{i+n} = w_i + k$ for $i \geq 0$. We call $w_0, w_1, \cdots$ the weight parameters of A with respect to k, p and n.

Let $A_i = a_{i+1} \cdots a_{i+n}$ for $i \geq 0$. In particular, $A_0 = A$. We note that

$$w_i = a_{i+1} + \cdots + a_{i+n} - k = w(A_i) - k \text{ for } i \geq 0. \tag{5.2}$$

We also note that $w(A_i) = w_i + k$ for $i \geq 0$.

Since $k \leq w(A) \leq k + p + 1$, then by Part 6 in [1] we get that

$$k \leq w(A_i) \leq k + p + 1 \text{ and } 0 \leq w_i \leq p + 1 \text{ for } i \geq 0. \tag{5.3}$$

6. The main algorithm.

Suppose k, p, n and $A = a_1 \cdots a_n$ where $k \le w(A) \le k + p + 1$.

Algorithm 6.1. Let $A_0 = A = a_1 \cdots a_n$ and $w_0 = a_1 + \cdots + a_n - k$.
Suppose $A_i = a_{i+1} \cdots a_{i+n}$ and w_i are calculated where $i \ge 0$.
1) If $w_i = 0$ and $a_{i+1} = 1$, then $a_{n+i+1} = 1$ and $w_{i+1} = w_i = 0$.
2) If $0 < w_i \le p + 1$ and $a_{i+1} = 1$, then $a_{n+i+1} = 0$ and $w_{i+1} = w_i - 1$.
3) If $w_i = p + 1$ and $a_{i+1} = 0$, then $a_{n+i+1} = 0$ and $w_{i+1} = w_i = p + 1$.
4) If $0 \le w_i < p + 1$ and $a_{i+1} = 0$, then $a_{n+i+1} = 1$ and $w_{i+1} = w_i + 1$.
5) Let $A_{i+1} = a_{(i+1)+1} \cdots a_{(i+1)+n} = a_{i+2} \cdots a_{i+1+n}$.
A stop criterium must be included.

In Part 6 of [1] it is proved that $w_0, w_1, w_2, \cdots$ are the weight parameters of A with respect to k, p and n, and $a_1 a_2 \cdots$ is generated from A by the symmetric shift register with parameters k, p and n.

Example 6.2. $A = a_1 \cdots a_{13} = 1110000111101$, $k = 6$, $p = 2$ and $n = 13$. Let $A_0 = A$. We note that $w(A_0) = 8$ and $w_0 = w(A_0) - k = 8 - 6 = 2$ etc. Then we get that

$$\begin{aligned}
A_0 &= a_1 \cdots a_{13} = 1110000111101 \quad w(A_0) = 8 \text{ and } w_0 = 2\\
A_1 &= a_2 \cdots a_{14} = 1100001111010 \quad w(A_1) = 7 \text{ and } w_1 = 1\\
A_2 &= a_3 \cdots a_{15} = 1000011110100 \quad w(A_2) = 6 \text{ and } w_2 = 0\\
A_3 &= a_4 \cdots a_{16} = 0000111101001 \quad w(A_3) = 6 \text{ and } w_3 = 0\\
A_4 &= a_5 \cdots a_{17} = 0001111010011 \quad w(A_4) = 7 \text{ and } w_4 = 1\\
A_5 &= a_6 \cdots a_{18} = 0011110100111 \quad w(A_5) = 8 \text{ and } w_5 = 2\\
A_6 &= a_7 \cdots a_{19} = 0111101001111 \quad w(A_6) = 9 \text{ and } w_6 = 3\\
A_7 &= a_8 \cdots a_{20} = 1111010011110 \quad w(A_7) = 9 \text{ and } w_7 = 3\\
A_8 &= a_9 \cdots a_{21} = 1110100111100 \quad w(A_8) = 8 \text{ and } w_8 = 2\\
A_9 &= a_{10} \cdots a_{22} = 1101001111000 \quad w(A_9) = 7 \text{ and } w_9 = 1\\
A_{10} &= a_{11} \cdots a_{23} = 1010011110000 \quad w(A_{10}) = 6 \text{ and } w_{10} = 0, \text{ etc.}
\end{aligned}$$

7. Determination of the minimal period.

Suppose $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ is generated from $A = a_1 \cdots a_n$ by the symmetric shift register with parameters k, p and n satisfying

$$0 \le k \le w(A) \le k + p + 1 \le n.$$

Use the algorithm in Section 6 to calculate $A_0, A_1, \cdots$ until $A_r = A$. Then r is the minimal period of $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$. That means, r is

the minimal period of A with respect to the symmetric shift register with parameters k, p and n.

Example 7.1. We suppose $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ is generated from $A = 11100001100001$ by he symmetric shift register with parameters $k = 3$, $p = 2$ and $n = 14$. Then $Q = (3, 4, 2, 4, 1, 0)$ is the even vector representation of A since $A = 1_3 0_4 1_2 0_4 1_1 0_0$.
In this case we get by the direct method that 982 is the minimal period of A^∞. That means, 982 is the minimal period of of A with respect to the symmetric shift register with parameters k, p and n.

Example 7.2. Suppose $Q = (2, 2, 100, 3, 100, 3, 100, 4)$, $k = 300$ and $p = 10$. Then Q represents the string $A = 1_2 0_2 1_{100} 0_3 1_{100} 0_3 1_{100} 0_4$ and $n = 314$.
Suppose $A^\infty = a_1 \cdots a_n a_{n+1} a_{n+2} \cdots$ is generated from A by he symmetric shift register with parameters k, p and n.
In this case we get by the direct method that 7 083 892 is the minimal period of A^∞. That means, 982 is the minimal period of of A with respect to the symmetric shift register with parameters k, p and n.

If the periods are very large, then the direct approach can be too tedious. In this case we can use the indirect method in menu option 3. We will not specify what we mean by "very large". This depends on the computer and how much time we will use. However, for example period sizes of 5 - 10 million can be calculated within a few seconds on most computers by the direct method.